New-4

MEMBERSHIP

Stars in catalog	29
with Gaia GSP-Phot	14 (48%)
with radial velocity	8
Stream id (sID)	18

GEOMETRY · great-circle fit

Centroid RA, Dec	90.3, -38.3 deg
Centroid l, b	243.4, -27.6 deg
Pole RA, Dec	47.3, 44.0 deg
Length (φ1, 2–98%)	17.0 deg
Width σ(φ2), robust	0.09 deg

KINEMATICS

pmRA* med ± σ	3.80 ± 0.31 mas/yr
pmDE med ± σ	0.04 ± 0.11 mas/yr
μφ1* med ± σ	3.36 ± 0.19 mas/yr
μφ2 med ± σ	-1.78 ± 0.45 mas/yr
Parallax (cat.) med	0.067 ± 0.117 mas
Parallax (wtd, ZP-corr.)	0.075 ± 0.021 mas
Distance	d = 13.4 ± 3.8 kpc
Physical length	3.97 kpc
RV median ± σ(MAD)	95.7 ± 1.7 km/s
RV gradient dV/dφ1	0.15 km/s/deg

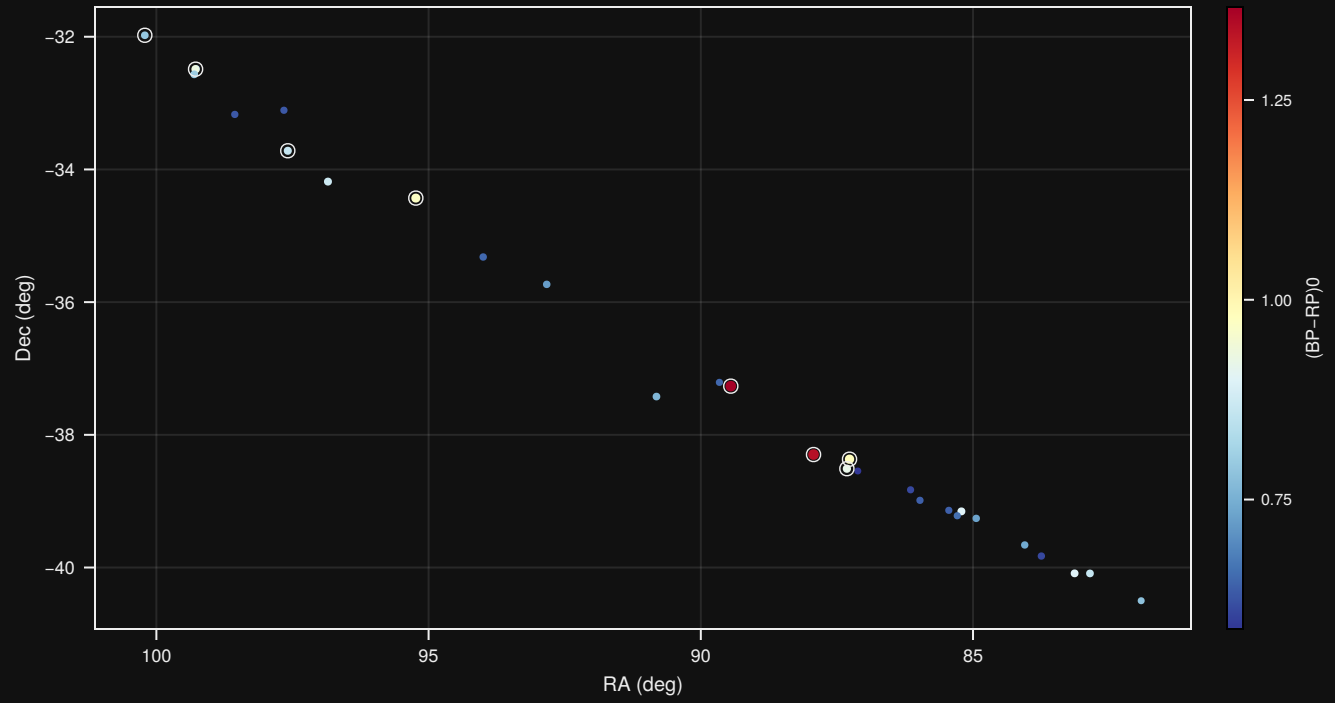
PHOTOMETRY · POPULATIONS

G0 range (median)	12.9–19.6 (18.7)
(BP−RP)0 median	0.76
GSP-Phot [M/H] (n=14)	-1.73 ± 0.46
GSP-Phot Teff median	5343 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (12 stars)

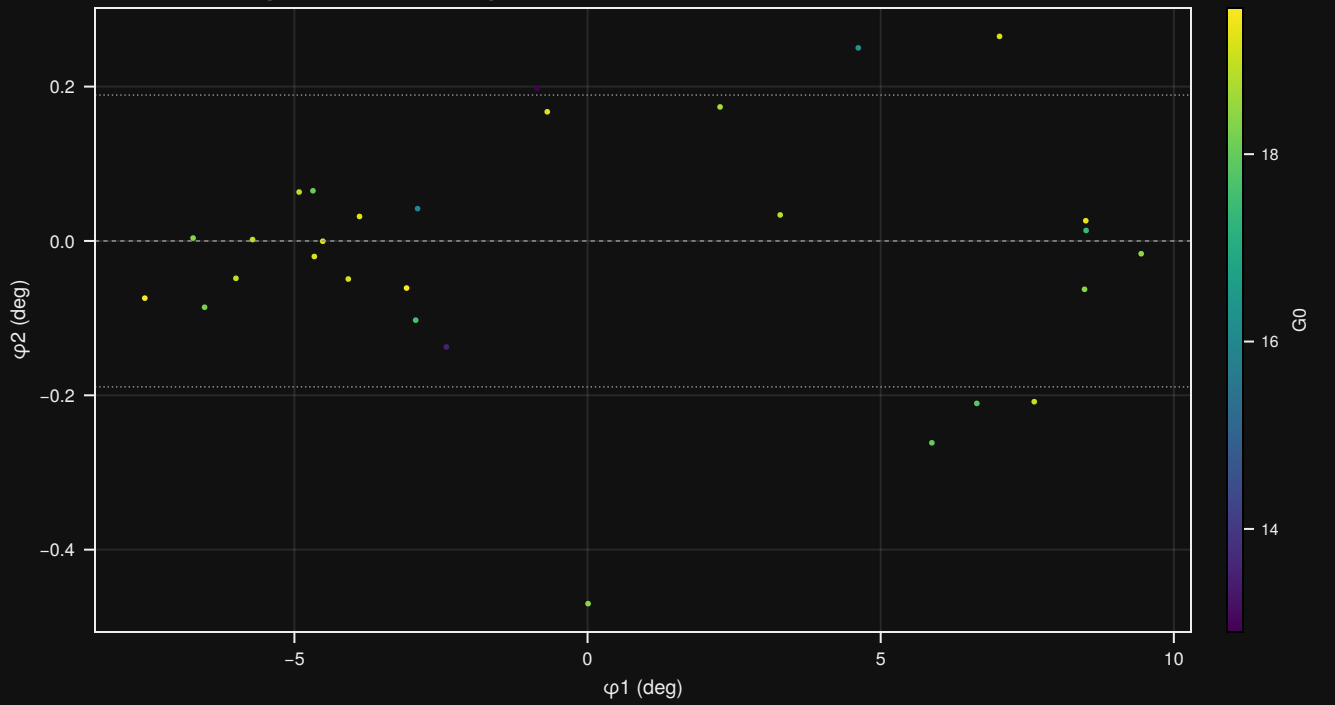
IBATA+2024 · TABLE 3

n catalog | paper	29 | 29
n_RV catalog | paper	8 | 8
<[Fe/H]> paper	-1.76
Stellar mass (paper)	730 Msun
Pole offset vs paper	3.3 deg

Sky map · catalog members (n = 29) · size ∝ brightness · ring = has RV

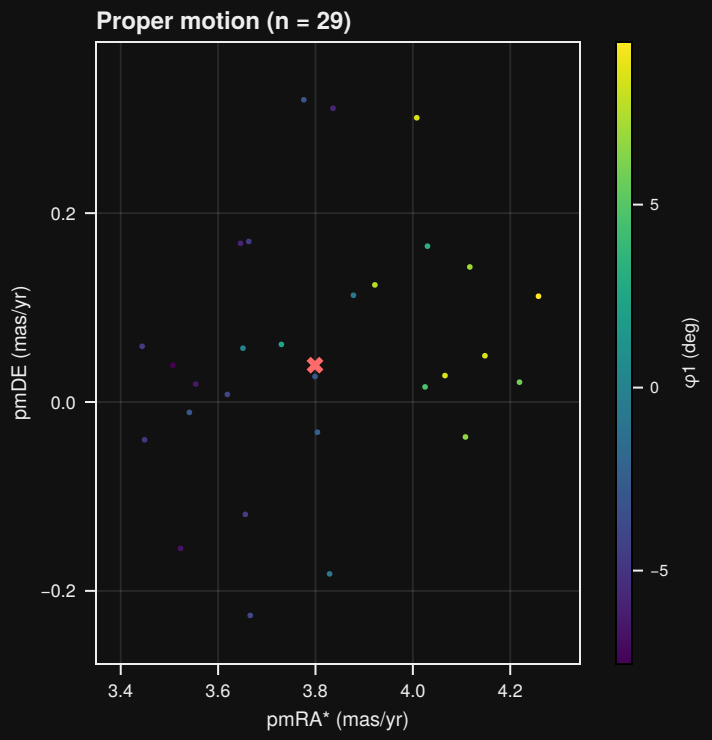


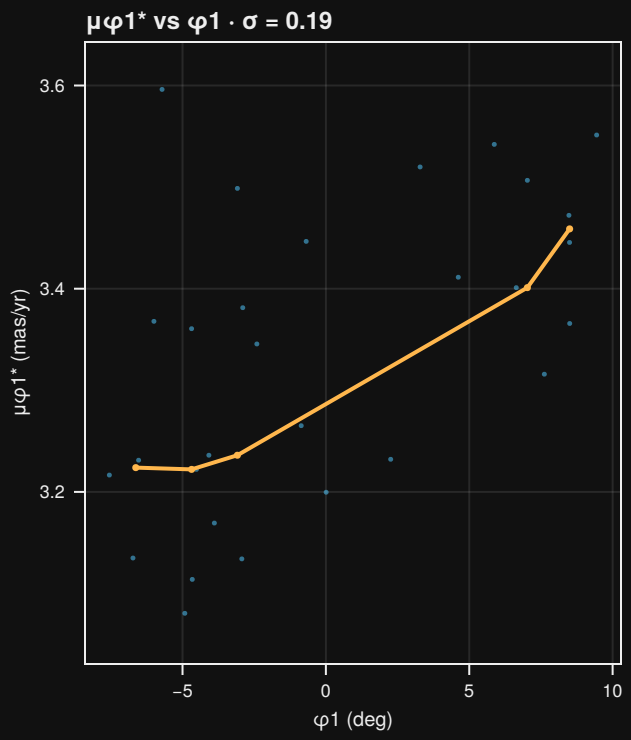
Stream frame (great-circle fit) · length 17.0° · width σ 0.09°

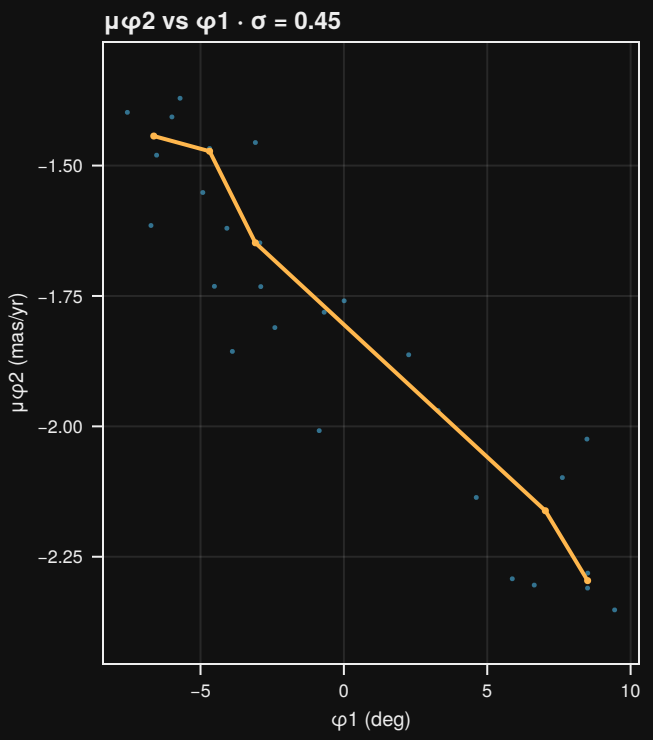


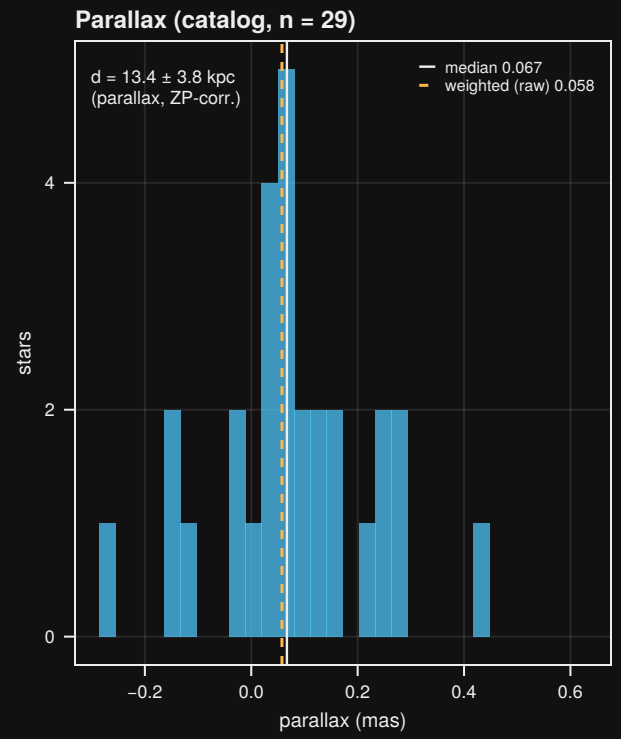
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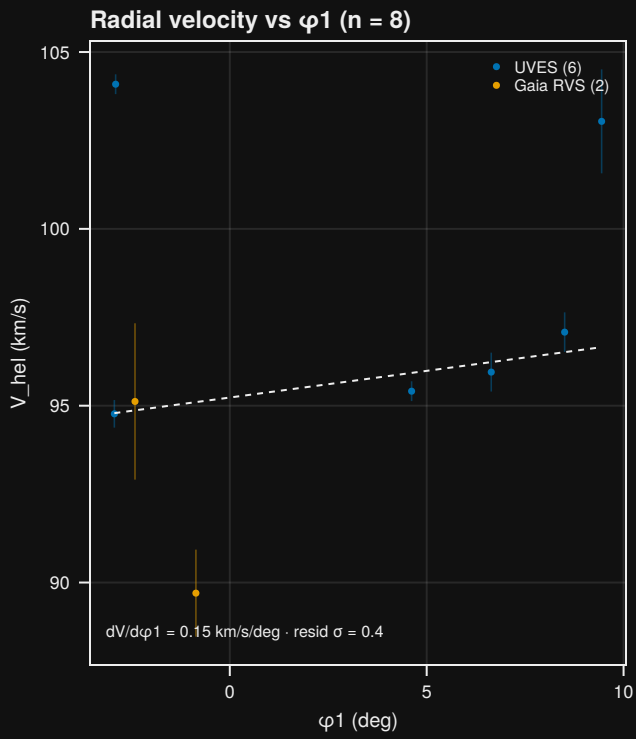
Kinematics · proper motion · parallax · radial velocity

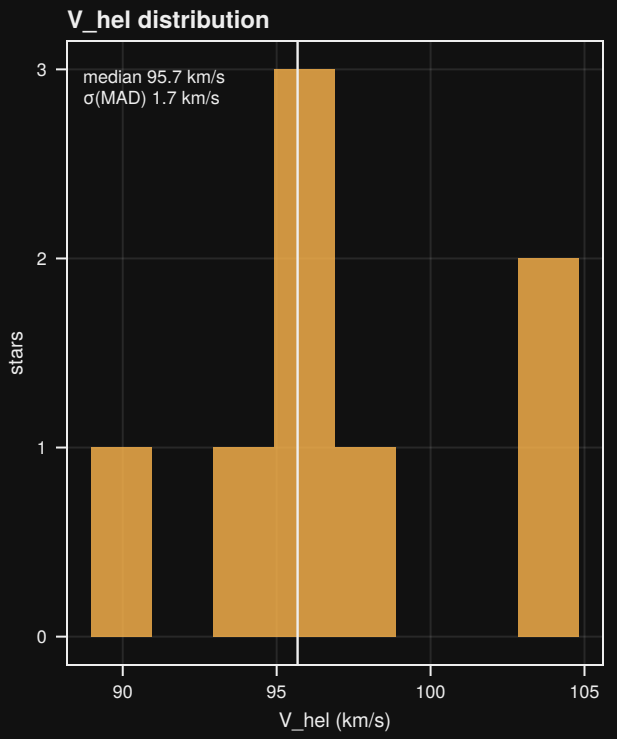








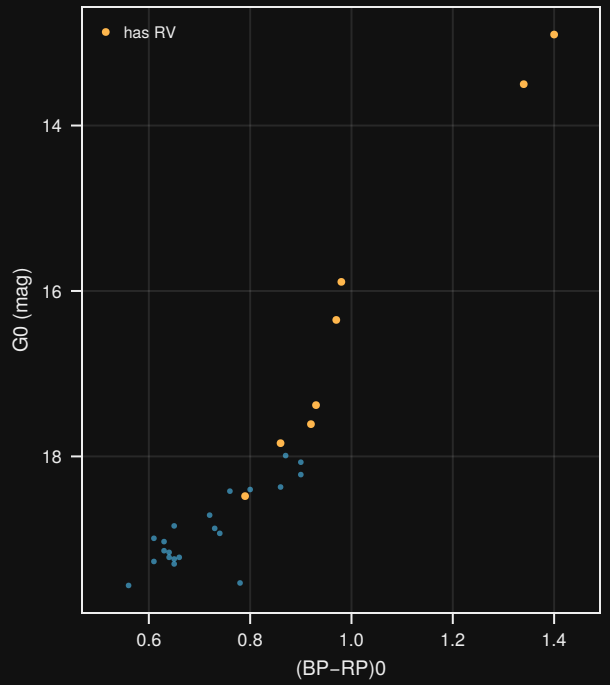




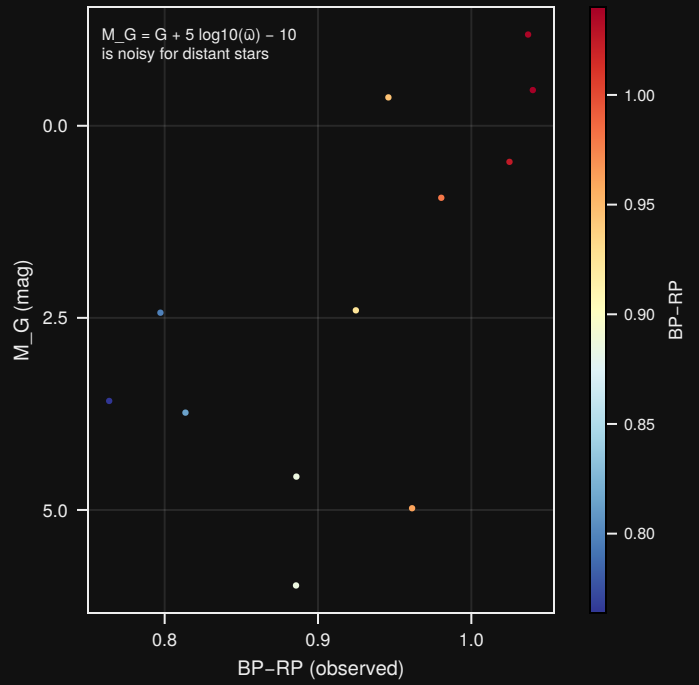
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Photometry · stellar types

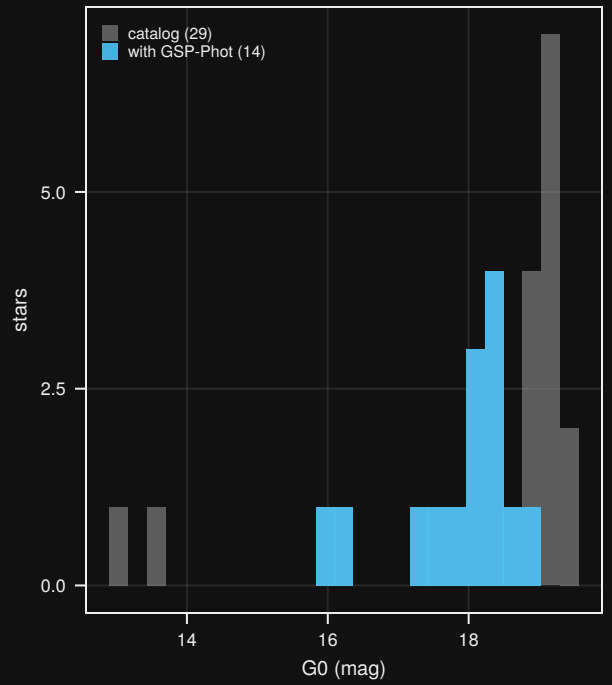
CMD · dereddened (n = 29)



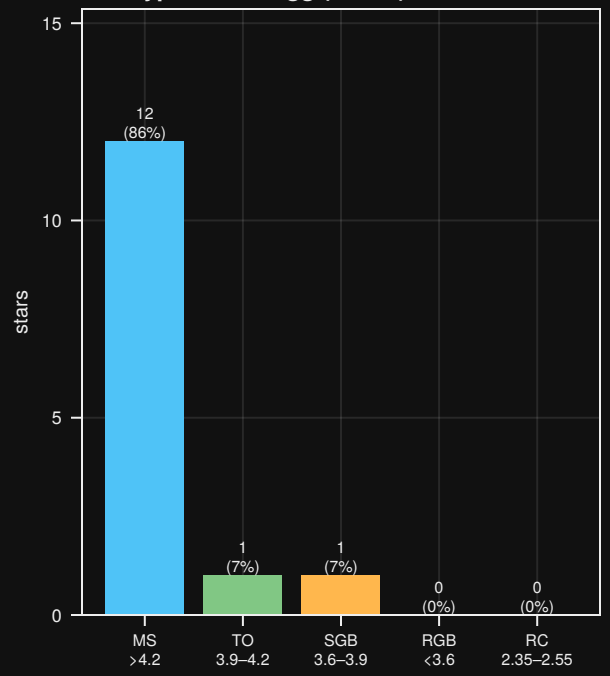
HR diagram · ϖ>0 (n = 12/14)



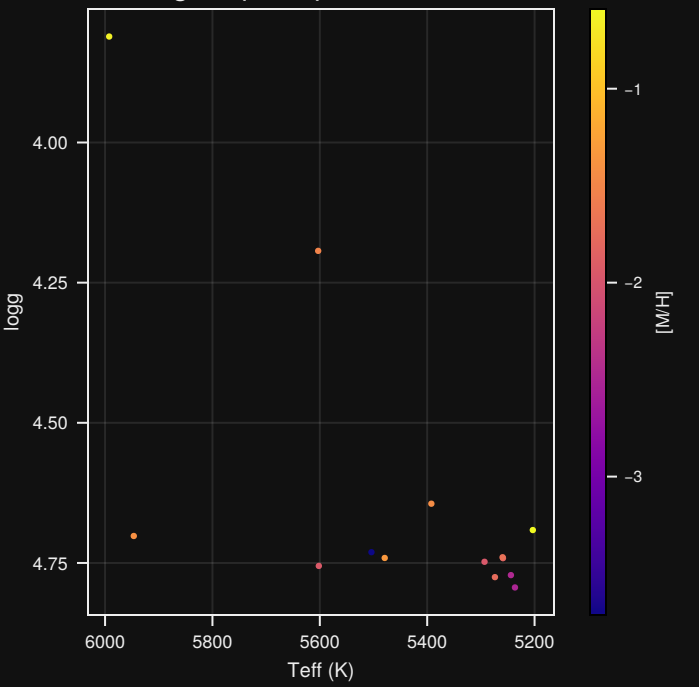
G0 & GSP-Phot coverage



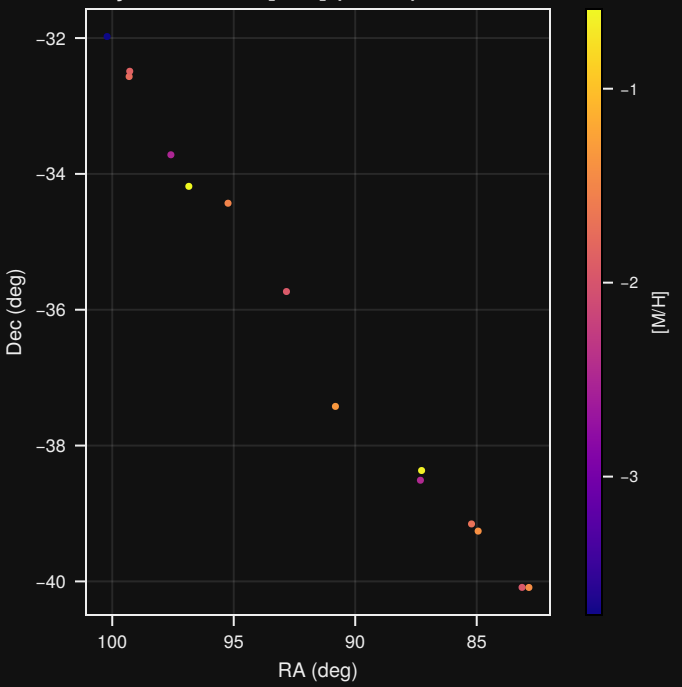
Star types from logg (n = 14)



Kiel diagram (n = 14)



Sky · GSP-Phot [M/H] (n = 14)

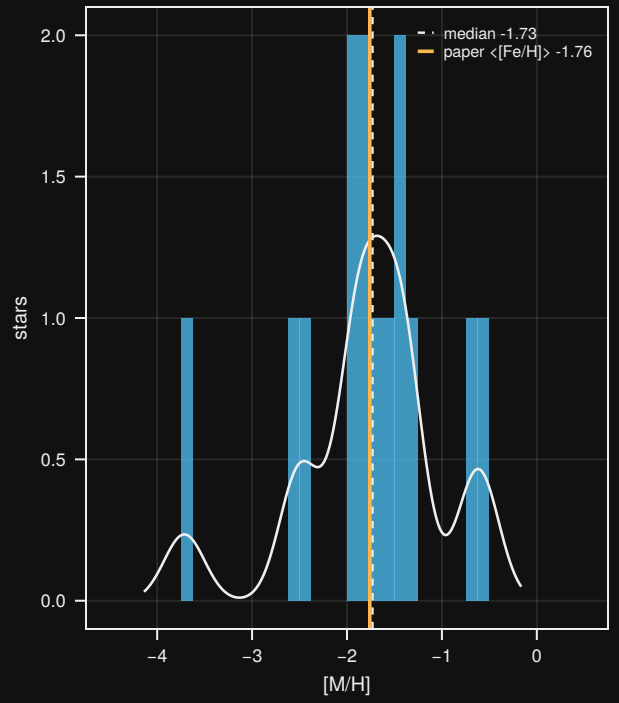


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

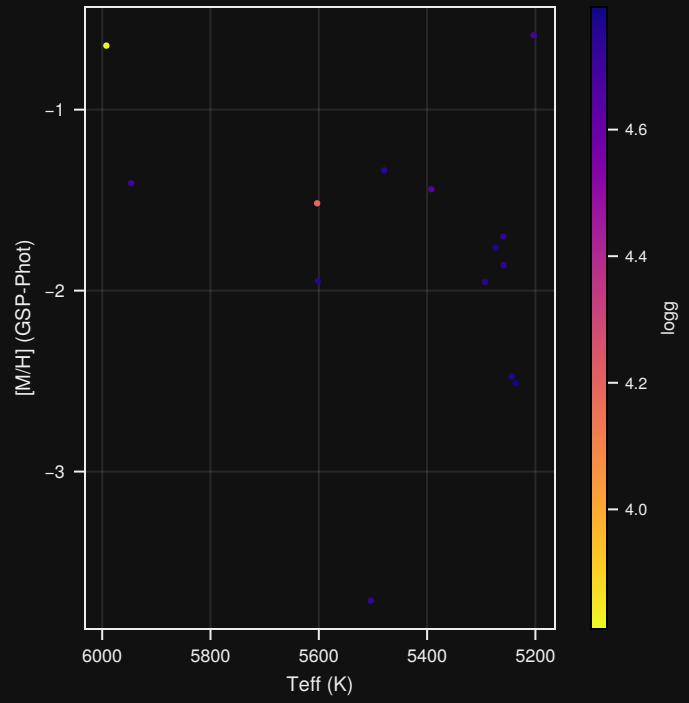
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Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 14)



Teff vs [M/H] (n = 14)



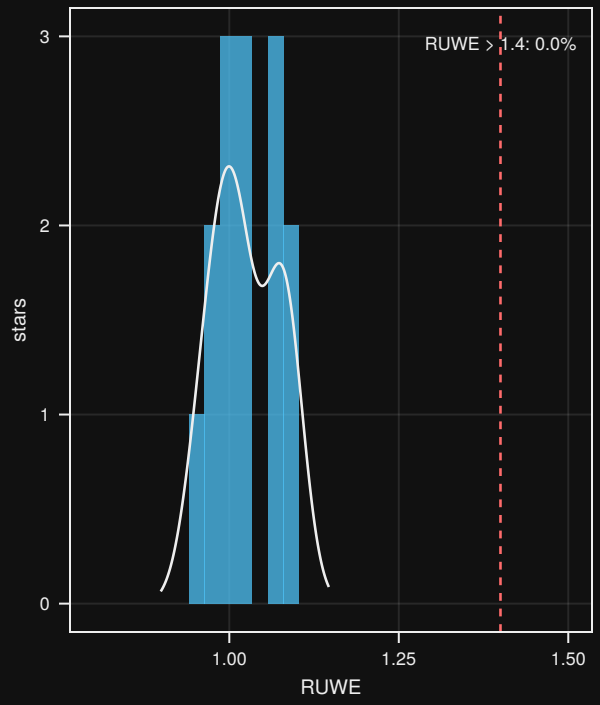
FLAME age (n = 2)

FLAME ages for only 2 star(s)
(FLAME needs good parallaxes)

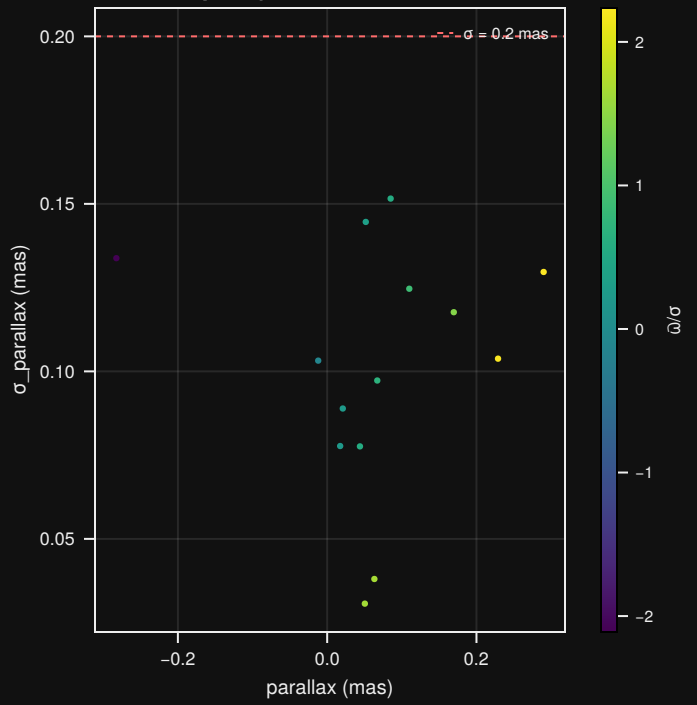
Age–metallicity (n = 2)

n < 3

RUWE (n = 14)

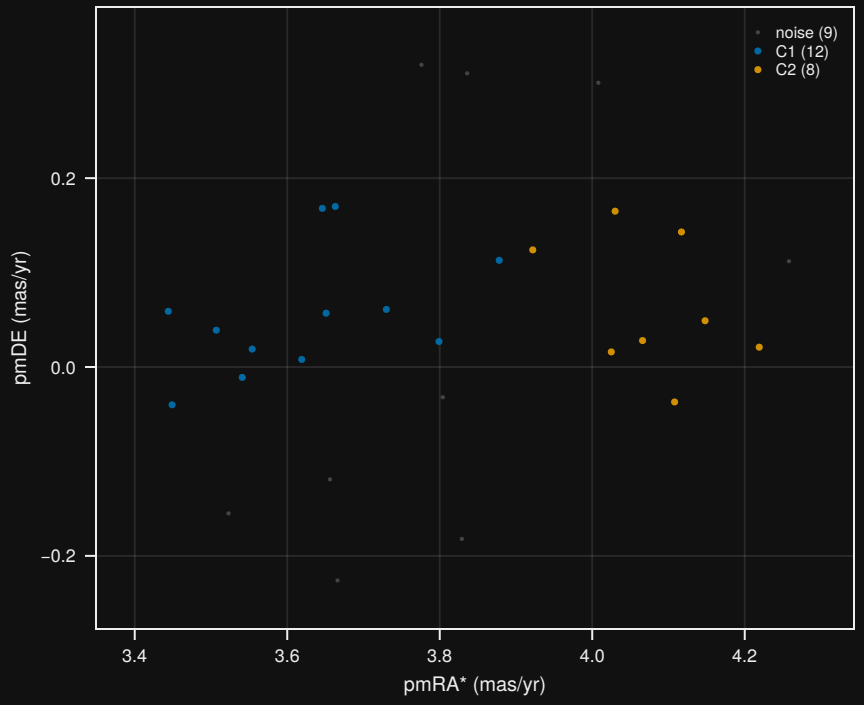


Parallax quality (n = 14)

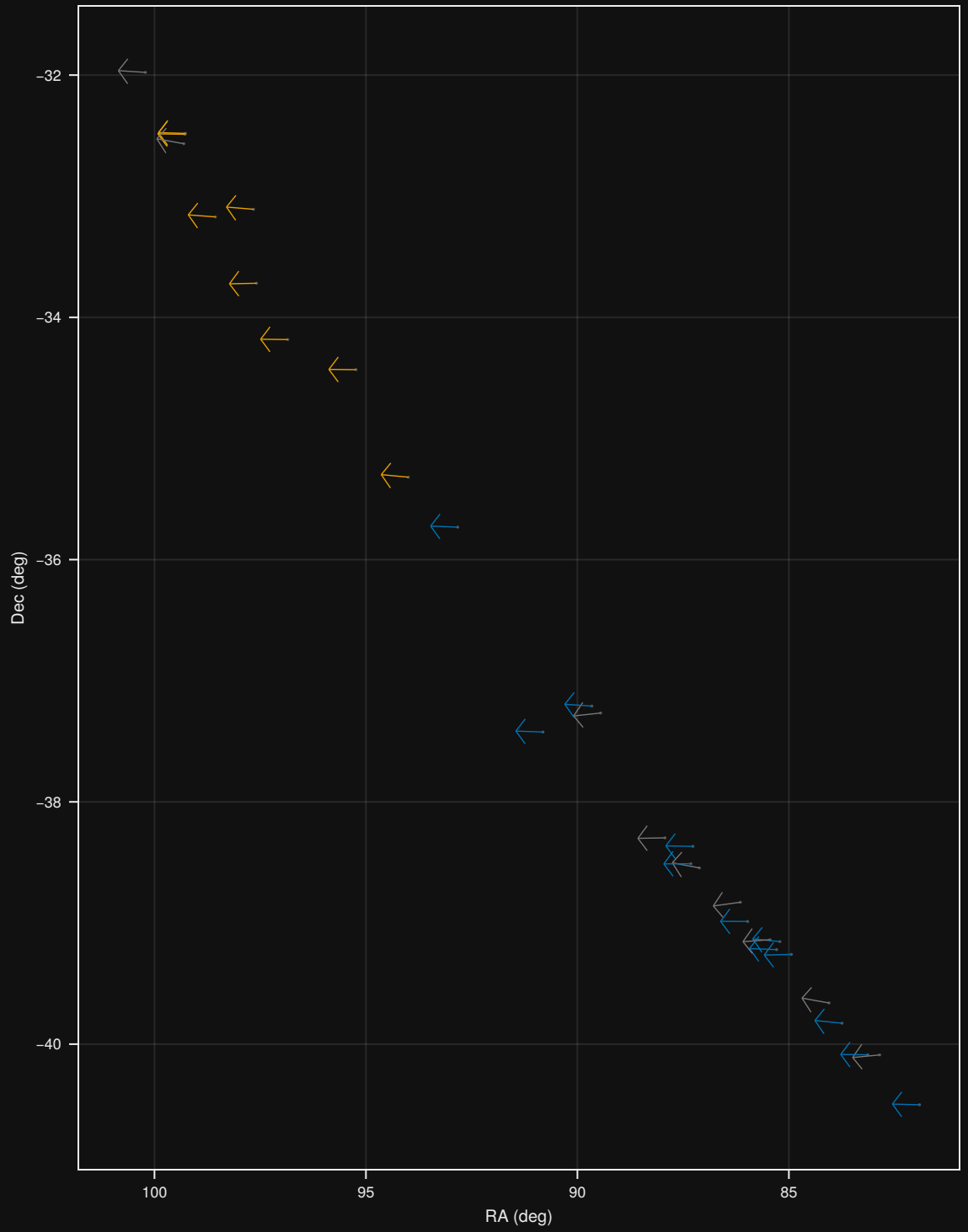


New-4

PM clustering · HDBSCAN · 2 cluster(s) · noise 31%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	9	3.80	-0.03	0.20	-1.76	3
C1	12	3.63	0.05	0.14	-1.70	2
C2	8	4.09	0.04	0.09	-1.69	3

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Per-cluster proper-motion vector maps

